doors are to be located within 15 m of the turnstiles.

The exception allows for the option to provide security access turnstiles that more closely resemble the operation of doors or gates [see Commentary Figure 1010.5.2 and Figure 1010.5.3(2)]. The operation of these types allows for free egress, so an extra door is not needed.

## SECTION 1011 STAIRWAYS

**1011.1 General.** *Stairways* serving occupied portions of a building shall comply with the requirements of Sections 1011.2 through 1011.13 . *Alternating tread devices* shall comply with Section 1011.14 . Ship's ladders shall comply with Section 1011.15 . Ladders shall comply with Section 1011.16 .

**Exception:** Within rooms or spaces used for assembly purposes, stepped aisles shall comply with Section 1030.

❖ It is important for stairway safety that all stairways meet the provisions in this section. This would include all elevation changes using stairways; everything from one riser to multiple flights and landings between stories (see the definitions for "Stair" and "Stairway" in Chapter 2). These provisions will be applicable for interior exit access stairways, interior exit stairways and exterior exit stairways, as well as any steps along the paths for exit access or exit discharge. It is intended that this section be applicable to required stairways as well as what can be called "convenience" stairways.

It is important to understand the terminology. Exit stairways are stairways that provide a protected path of egress travel between the exit access and the exit discharge. Interior exit stairways are required to be enclosed in accordance with Section 1023. Exterior exit stairways are protected by the exterior wall of the building and must comply with Section 1027. Exit access stairways are typically unenclosed interior stairways and comply with Section 1019 where they provide access between stories. Exit access travel distance stops at an exit stairway enclosure, but includes any travel down an exit access stairway. Stairways that are outside and provide a route from the level of exit discharge to grade are considered part of the exit discharge. See the commentary in Chapter 2 for the defined term "Exit discharge, level of."

Sections 1011.2 through 1011.13 provide criteria for the typical stairway. Special provisions are provided for curved stairways (Section 1011.9 ), spiral stairways (Section 1011.10 ), and stairways to the roof for fire department access and for elevator equipment service (Sections 1011.12 and 1011.12.1 ).

Items that provide vertical access similar to stairways are also addressed in this section: alternating tread devices (Section 1011.14 ), ship's ladders (Section 1011.15 ) and ladders (Section 1011.16 ). These devices are only permitted to provide access to very limited spaces.

The exception indicates that stepped aisles (previously called aisle stairs) are addressed in Section 1030. Having this exception at the beginning of the stairway section negates the need for repeated exceptions throughout the stairway provisions. While both stairways and stepped aisles are a series of treads and risers, how occupants move on and off and configurations are very different. Occupants leave and join stepped aisles along the entire run, while occupants only enter stairways at the top and bottom. Stepped aisles have center handrails with breaks to allow for access into seating, while most stairways have handrails on both sides. Stepped aisles can be nonuniform in some locations to allow for parabolic seating bowls, while uniform tread and riser configurations are required for stairways. Section 1030 should be used for stepped aisles between and immediately adjacent to seating or where the steps are a direct continuation of the stepped aisles and lead to a level cross aisle or floor. Section 1011 is used for stairways that lead from the balcony, concourse or cross aisle to a floor level above or below the seating areas (see Sections 1011.5.2, 1011.5.4 and 1011.6).

See Sections 1008.2, 1008.2.1 and 1008.3 for stairway lighting requirements.

**1011.2 Width and capacity.** The required capacity of *stairways* shall be determined as specified in Section 1005.1 , but the minimum width shall be not less than 1100 mm. See Section 1009.3 for accessible *means of egress stairways*.

**Exceptions**:

1 *Stairways* serving an *occupant load* of less than 50 shall have a width of not less than 900 mm.

2 *Spiral stairways* as provided for in Section 1011.10 .

3 Where an incline platform lift or stairway chairlift is installed on *stairways* serving occupancies in Group R-3, or within *dwelling units* in occupancies in Group R-2, a clear

passage width not less than 500 mm shall be provided. Where the seat and platform can be folded when not in use, the distance shall be measured from the folded position.

❖ To provide adequate space for occupants traveling in opposite directions and to permit the intended full egress capacity to be developed, minimum dimensions are dictated for means of egress stairways. A minimum width of 1100 mm is required for stairway construction to permit two columns of users to travel in the same or opposite directions. The reference to Section 1005.1 is for the determination of stairway width based on the occupant load it will serve (i.e, capacity). The larger of the two widths is to be used.

Exception 1 recognizes that relatively small occupant loads of less than 50 permit a staggered file of users where traveling in the same direction. Where traveling in opposite directions, one column of users must stop their ascent (or descent) to permit the opposite column to continue. Again, considering the relatively small occupant loads, any disruption of orderly flow will be infrequent and brief. The use of this exception is limited to buildings where the entire occupant load of each upper story and/or basement is less than 50.

Exception 2 permits a spiral stairway to have a minimum width of 660 mm where it conforms to Section 1011.10 , on the basis that the configuration of a spiral stairway will allow nothing other than single-file travel.

Exception 3 addresses the use of inclined platform lifts or stairway chairlifts for individual dwelling units. For clarification on the types of lifts, see the commentary to Section 1110.9 . Both types of lifts may be installed to aid persons with mobility impairments in their homes. The code and ASME A18.1 allow for a reduction in the width of the stair to a minimum of 500 mm of clear passageway to be maintained on a stairway where a lift is located. If a portion of the lift, such as a platform or seat, can be folded, the minimum clear dimension is to be measured from the folded position. If the lift cannot be folded, then the 500 mm is measured from the fixed position. The track for these lifts typically extends 230 to 300 mm from the wall, making the 500 mm clear measurement actually 600 to 690 mm from the edge of the track.

The code does not have any specific provisions for where incline platform lifts are utilized along stairways in locations other than within dwelling units. Section 1110.9 limits the use of platform lifts in new construction to mainly areas with minimal occupant loads or where elevators and ramps are impracticable. SBC 901 Sections 306.7.8 allows for platform lifts anywhere in existing buildings in order to gain accessibility for persons with mobility impairments. When in the closed and off position, platform lifts should not block the clear width required for the stairway or use of the handrails. The industry is currently working on different options to address the concern that the lift may be in operation during an event that requires evacuation.

**1011.3 Headroom.** *Stairways* shall have a headroom clearance of not less than 2100 mm measured vertically from a line connecting the edge of the *nosings*. Such headroom shall be continuous above the *stairway* to the point where the line intersects the landing below, one tread depth beyond the bottom riser. The minimum clearance shall be maintained the full width of the *stairway* and landing.

**Exceptions**:

1 *Spiral stairways* complying with Section 1011.10 are permitted a 2000 mm headroom clearance.

2 In Group R-3 occupancies; within *dwelling units* in Group R-2 occupancies; and in Group U occupancies that are accessory to a Group R-3 occupancy or accessory to individual *dwelling units* in Group R-2 occupancies; where the *nosings* of treads at the side of a *flight* extend under the edge of a floor opening through which the *stair* passes, the floor opening shall be allowed to project horizontally into the required headroom not more than 120 mm.

❖ This headroom requirement is necessary to avoid an obstruction to orderly flow and to provide visibility to the users so that the desired path of travel can be planned and negotiated. Height is a vertical measurement above every point along the stairway stepping and walking surfaces, with minimum height measured vertically from the tread nosing or from the surface of a landing or platform up to the ceiling [see Commentary Figure 1011.3(1)].

Sections 1003.2 and 1207.2 require a minimum ceiling height within a room of 2300 mm. A bulkhead or doorway at the bottom of the stairway would be allowed to meet the minimum headroom height of 2000 mm, as permitted in Section 1003.3 .

Exception 1, allowing for a clear headroom of 2000 mm for spiral stairs, correlates with the provisions of Section 1011.10 .

Exception 2 recognizes a common method of stair-well construction in which the stringer on the open side of a stair is supported by the same floor joists or wall that supports the edge of the opening through which the stairway passes to the floor above, thus resulting in the stairway being wider at the lower portion than at the top portion. In this case, headroom is not required for a distance of up to 120 mm measured horizontally from the edge of the opening above to the handrail or guard system, which limits the clear width on the lower open sides of the stairway. The 120 mm maximum is derived from the finished width of a typical 2 by 4 supporting wall and is not critical to obstructing orderly flow or visibility in the desired path of travel [see Commentary Figure 1011.3(2)].

**1011.4 Walkline.** The walkline across *winder* treads shall be concentric to the direction of travel through the turn and located 300 mm from the side where the *winders* are narrower. The 300 mm dimension shall be measured from the widest point of the clear *stair* width at the walking surface of the *winder*. Where *winders* are adjacent within the *flight*, the point of the widest clear *stair* width of the adjacent *winders* shall be used.

❖ This requirement is essential for smooth, consistent travel on stairs that turn with winder treads. It provides a standard location for the regulation of the uniform tread depth of winders. Because of the wide range of anthropometrics of stairway users, there is no one line that all persons will travel on stairs; however, the code recognizes a standard location of a walkline is essential to design and enforcement. Each footfall of the user through the turn can be associated with an arc to describe the path traveled. As a user ascends or descends the flight, the turning at each step should be consistent through the turn. The walkline is established concentric to or having the same center (approximately parallel) as, the arc of travel of the user. The tread depth dimension at the walkline is one of two tread depths across the width of the stair at which winder tread depth is regulated, cited in Section 1011.5.2 . The second is the minimum tread depth. Regulation at these two points controls the angularity of the turn and the configuration of the flight. In order to establish consistently shaped winders, tread depths must always be measured concentric to the arc of travel. The walkline is unique as the only line or path of travel where winder tread depth is controlled by the same minimum tread depth as rectangular treads. However, Exception 2 of Section 1011.5.4 recognizes that winder tread depth need not be compared to rectangular tread depths for dimensional uniformity in the same flight because the location of the walkline is chosen for the purpose of providing a standard and cannot be specific to the variety of actual paths followed by all users. This specific line location is determined by measuring along each nosing edge 300 mm from the extreme of the clear width of the stair at the surface of the winder tread or the limit of where the foot might be placed in use of the stair. If adjacent winders are present, the point of the widest clear stair width at the surface of the tread in the group of adjacent consecutive winders is used to provide the reference from which the 300 mm dimension will be measured along each nosing. The tread depth may be determined by measuring between adjacent nosings at these determined intersections of the nosings with the walkline. It is important to note that the clear stair width is only that portion of the stair width that is clear for passage. Portions of the stair beyond the clear width are not consequential to use of the stair, consistent travel or location of the walkline.

**1011.5 Stair treads and risers.** *Stair* treads and risers shall comply with Sections 1011.5.1 through 1011.5.5.3 .

❖ The provisions for treads and risers contribute to the efficient use of the stairway, facilitating smooth and consistent travel. This section provides dimensional ranges and tolerances for the component elements to allow the flexibility required to design and construct a stair or a flight of stairs that are elements of a stairway. The allowed proportion of maximum riser height and minimum tread depth provide for a maximum angle of ascent, but there is no maximum tread depth to consider with the minimum riser height that would define a minimum angle for a stairway. Nor is the proportion of riser height to tread depth compared with the limitations of the length of the user's stride on stairways, which is significantly foreshortened from the user's stride on the level. For this reason, care should be taken where incorporating larger tread depths and controlling the point at which a tread might be wide enough to require more than one step to cross, which can vary significantly where considering ascent and descent movement patterns. Especially in areas where all segments of the public might use the stairs, those persons requiring two smaller sequential steps to cross the tread would progress at significantly different rates

than those who might be able to stretch or jump, which could lead to dangerous complications, especially in egress. Of equal significance is the use of shorter risers without increasing tread depth, resulting in a proportion that could cause overstepping. However, by controlling the minimum depth of rectangular treads and the minimum depth and angularity of winder treads, these components can dictate the configuration of the plan for a flight of stairs to provide for smooth and consistent travel.

Section 1011.5.1 provides for consistent identification of the surfaces that are to be measured. This is as critical to the users' experience throughout the built environment as it is to determining all the dimensions in this section as they must relate to each other and the ultimate design and construction of a safe stairway.

The remaining sections address the uniformity and essential attributes of the tread and riser. Of particular note is the nosing, or leading edge, of the tread. The nosing shape and projection affect the determination of the tread depth and riser height and are regulated for all steps throughout the stairway, including the nosings at landings, to provide for smooth and consistent travel. Furthermore, the line connecting the nosings is used to determine handrail height, guard height and headroom, making consistent nosings integral to every element of the stairway design, construction and regulation.

**1011.5.1 Dimension reference surfaces.** For the purpose of this section, all dimensions are exclusive of carpets, rugs or runners.

❖ Carpets, rugs and runners, like furniture, are frequently changed by the occupants and are not regulated by the code. For this reason, it is essential that the riser height and tread depth be regulated exclusive of these transitory surfaces to provide an enforceable standard. This practice minimizes possible variation due to the removal of nonpermanent carpeting throughout the life of a structure and provides a standard enforcement methodology that will provide consistency across the built environment for all users. Where owners or occupants add carpeting, rugs or runners, they need to add it to all tread and landing surfaces in the stairway. It is important that the tread and landing surfaces are consistent and comply with the code prior to the addition of carpet. This methodology of enforcement makes it unnecessary to reconstruct floor and stair elevations in the stairway where nonpermanent carpet surfaces that do not require a building permit are changed and eliminates the resulting variations in the built environment that would not comply with the tolerance in Section 1011.5.4 (see Commentary Figure 1011.5.1).

**1011.5.2 Riser height and tread depth.** *Stair* riser heights shall be 175 mm maximum and 100 mm minimum. The riser height shall be measured vertically between the *nosings* of adjacent treads or between the stairway landing and the adjacent tread. Rectangular tread depths shall be 275 mm minimum measured horizontally between the vertical planes of the foremost projection of adjacent treads and at a right angle to the tread's *nosing*. *Winder* treads shall have a minimum tread depth of 275 mm between the vertical planes of the foremost projection of adjacent treads at the intersections with the walkline and a minimum tread depth of 250 mm within the clear width of the *stair*.

**Exceptions:**

1. *Spiral stairways* in accordance with Section 1011.10 .
2. *Stairways* connecting stepped *aisles* to cross *aisles* or concourses shall be permitted to use the riser/ tread dimension in Section 1030.14.2 .
3. In Group R-3 occupancies; within *dwelling units* in Group R-2 occupancies; and in Group U occupancies that are accessory to a Group R-3 occupancy or accessory to individual *dwelling units* in Group R-2 occupancies; the maximum riser height shall be 200 mm; the minimum tread depth shall be 250 mm; the minimum *winder* tread depth at the walkline shall be 250 mm; and the minimum *winder* tread depth shall be 150 mm. A *nosing* projection not less than 20 mm but not more than 32 mm shall be provided on *stairways* with solid risers where the tread depth is less than 275 mm.
4. See Section 503.1 of SBC *901* for the replacement of existing *stairways*.
5. In Group I-3 facilities, *stairways* providing access to guard towers, observation stations and control rooms, not more than 23 $m^2$ in area, shall be permitted to have a maximum riser height of 200 mm and a minimum tread depth of 225 mm.

❖ The riser height—the vertical dimension from tread surface to tread surface or tread surface to landing surface—is typically limited to not more than 175 mm or less than 100 mm. The minimum tread depth—the horizontal distance

from the leading edge (nosing) of one tread to the leading edge (nosing) of the next adjacent tread or landing— is typically limited to not less than 275 mm [see Commentary Figure 1011.5.2]. The minimum tread depth of 275 mm is intended to accommodate the largest shoe size found in 95 percent of the adult population, allowing for an appropriate overhang of the foot beyond the tread nosing while descending a stairway. Tread depths under 275 mm could cause a larger overhang (depending on the size of the foot) and could force users with larger feet to increase the angle of their foot to the line of travel while descending a stairway. Based on the probability of adequate foot placement, the rate of misstep with various step sizes and consideration for the user's comfort and energy expenditure, it was agreed that the 275 mm minimum tread depth and maximum 175 mm riser height resulted in the reasonable proportion of riser height and tread depth for stairway construction. A minimum riser height of 100 mm is considered to allow the visual identification of the presence of the riser in ascent or descent.

The precise location of rectangular tread depth and riser measurements is to be perpendicular to the tread's nosing or leading edge. This is to duplicate the user's anticipated foot placement in traveling the stairway.

The size for a winder tread is also considered for proper foot placement along the walkline [see Commentary Figure 1011.9 and the commentary for Section 1011.4 ]. The dimensional requirements are consistent with the straight tread.

The exceptions apply only to the extent of the text of each exception. For example, the entire text of Section 1011.5.2 is set aside for spiral stairways conforming to Section 1011.10 (see Exception 3). However, Exception 3 allows a different maximum riser and minimum tread under limited conditions but retains the minimum riser height and measurement method of Section 1011.5.2 .

Exception 1 is for spiral staircases, a unique type of stairway. Section 1011.5.2 is not applicable to this stair type because of construction issues and limited applications. For a discussion on spiral staircases, see Section 1011.10 .

Exception 2 provides a practical exception where assembly facilities are designed for viewing. See Section 1030.14.2 for assembly stepped aisles. This exception is limited to where stairways are a direct continuation of the path of travel from the level cross aisle to the stepped aisles. It is not permitted for other stairways within the assembly space.

Exception 3 allows revisions to the maximum 175 mm/275 riser/tread requirements for Group R-3 occupancies and any associated utility (such as barns, connected garages or detached garages) and within individual units of Group R-2 and their associated utility areas (such as attached garages). This change is allowed because of the low occupant load and the high degree of occupant familiarity with the stairways. Where this exception is applied for stairways that have solid risers, each tread is required to have a nosing projection with a minimum dimension of 20 mm and maximum dimension of 32 mm where the tread depth is less than 275 mm. Nosing projections are created where the nosing of the tread above extends beyond the trailing edge of the tread below or where a solid riser is angled under the tread above and connected to the trailing edge of the tread below. Nosing projections are not required for residential stairs with open risers and 250 mm treads. A nosing projection provides a greater stepping surface for those ascending the stairway. For users descending the stairway, the nosing projection allows the toe of the foot to be placed farther away from the riser above, providing the necessary clearance for the heel of the foot as it swings down in an arc to its position on the tread (see Commentary Figure 1011.5.3).

Exception 4 allows for the replacement of existing stairways. Where a change of occupancy would require compliance with current standards, this exception allows for a stairway that may be steeper than that permitted, provided it does not constitute a hazard (see SBC 901 Section 503.1).

In spaces of not more than 23 m² in correctional facilities (Group I-3), Exception 5 allows for steeper stairs with a maximum riser height of 200 mm and a minimum tread depth of 225 mm because of the minimal occupant load and the familiarity of the users with the stairway. Although not stated in this exception, utilizing a nosing projection to provide effective tread depth is a good design practice [see Exception 3 regarding tread depths less than 275 mm].

**1011.5.3 Winder treads.** Winder treads are not permitted in means of egress stairways except within a dwelling unit.

**Exceptions:**

1 Curved *stairways* in accordance with Section 1011.9 .

2 *Spiral stairways* in accordance with Section 1011.10 .

❖ This section is intended to coordinate the general provisions for stairway tread and riser dimensions in Section 1011.5.2 with the provisions for winder treads permitted in curved and spiral stairways (see Sections 1011.9 and 1011.10). Winders are permitted in means of egress stairways within dwelling units where occupant loads are small and occupants have increased familiarity (see Commentary Figure 1011.5.3 and Section 1011.5.2, Exception 3). This is consistent with provisions in the SBC 1101/1102.

Winders are used to change the direction of a flight by introducing a consistent incremental turn associated with each tread. The risk of injury in the use of stairways constructed with winders is considered to be greater than for stairways constructed as straight runs because users may be restricted by the presence of other users, limiting visual clues or influencing the rate of travel. Additional user attention in the turn and the aid of the turn in arresting falls (similar to turns at landings) is also understood to negate this.

The employment of winders in stairway construction may necessitate the change of the user's gait in both ascent and descent where the tread depth of the winder is not equal to the tread depth of any rectangular treads in the same flight. For example, a person descending a straight flight of stairs will develop a particular gait conforming to the proportion of the riser height and tread depth that will be consistent throughout the flight. However, in a flight that includes winders and rectangular treads, the user must accommodate a change in the proportion of the riser height and tread depth as determined by the path of travel chosen. Visual clues are important to the users' instinctive responses to alter the path of travel, the length of stride, or a combination of both that may result in nonconcentric movement. To ensure users of the visual clues necessary to alter their gait and limit the need to alter the path of travel in conditions of higher occupant loading, flights with winders must meet the specific safety provisions listed for curved or spiral stairways unless they are within a dwelling unit.

**1011.5.4 Dimensional uniformity.** *Stair* treads and risers shall be of uniform size and shape. The tolerance between the largest and smallest riser height or between the largest and smallest tread depth shall not exceed 10 mm in any *flight* of *stairs*. The greatest *winder* tread depth at the walkline within any *flight* of *stairs* shall not exceed the smallest by more than 10 mm.

**Exceptions**:

1 *Stairways* connecting stepped *aisles* to cross *aisles* or concourses shall be permitted to comply with the dimensional non-uniformity in Section 1030.14.2 .

2 Consistently shaped *winders*, complying with Section 1011.5 , differing from rectangular treads in the same *flight* of *stairs*.

3 Non-uniform riser dimension complying with Section 1011.5.4.1 .

❖ Dimensional uniformity in the design and construction of stairways contributes to safe stairway use. When ascending or descending a stair, users establish a gait based on the instinctive expectation or "feel" that each step taken will be at the same height and will land in approximately the same position on the tread as the previous steps in the pattern. A change in tread or riser dimensions in a stairway flight in excess of the allowed dimensional tolerance can break the rhythm and cause a misstep, stumbling or physical strain that may result in a fall or serious injury. Therefore, this section limits the dimensional variations to a tolerance of 10 mm between the largest and smallest riser or tread dimension in a flight of stairs. A "flight" of stairs is defined as a run of stairs between landings.

For special conditions of construction and as a practical matter, this section allows some greater variations in stairway tread and riser dimensions than the general limitations specified previously.

Exception 1 provides a practical exception where assembly facilities are designed for viewing. See Sections 1030.14 through 1030.14.2.4 for assembly stepped aisles. This exception is limited to where stairways are a direct continuation of the path of travel from the level cross aisle to the stepped aisles. It is not permitted for other stairways within the assembly space.

Exception 2 addresses winder treads, which must be consistent along the walkline (see Commentary Figure 1011.5.4) when compared to other winder treads in the same flight but are not required to meet the tolerance where compared to the uniform dimension of rectangular treads in the same flight.

Exception 3 is in recognition of the situation where a stairway meets a surface that slopes

up or down perpendicular to the stairway. See the commentary to Section 1105.4.1.

**1011.5.4.1 Nonuniform height risers.** Where the bottom or top riser adjoins a sloping *public way*, walkway or driveway having an established grade and serving as a landing, the bottom or top riser is permitted to be reduced along the slope to less than 100 mm in height, with the variation in height of the bottom or top riser not to exceed one unit vertical in 12 units horizontal (8-percent slope) of *stair* width. The *nosings* or leading edges of treads at such nonuniform height risers shall have a distinctive marking stripe, different from any other *nosing* marking provided on the *stair flight*. The distinctive marking stripe shall be visible in descent of the *stair* and shall have a slip-resistant surface. Marking stripes shall have a width of not less than 25 mm but not more than 50 mm.

❖ This section addresses the situation where the bottom riser of a flight of stairs meets a sloped landing, such as a public way, walk or driveway (see Commentary Figure 1011.5.4.1). Because the sidewalk landing is sloped, stepping off the bottom tread on one side will result in a higher riser than stepping off the bottom tread on the other side. This is permitted provided the bottom riser is marked so that someone using the stairs will be aware of the hazard of a nonuniform riser.

**1011.5.5 Nosing and riser profile.** *Nosings* shall have a curvature or bevel of not less than 2 mm but not more than 15 mm from the foremost projection of the tread. Risers shall be solid and vertical or sloped under the tread above from the underside of the *nosing* above at an angle not more than 30 degrees from the vertical.

❖ The profiles of treads and risers contribute to stairway safety. The radius or bevel of the nosing eases the otherwise square edge of the tread and prevents irregular chipping and wear that can become a maintenance issue and seriously affect the safe use of the stair. The minimum curvature or bevel of 2 mm eliminates a sharp square edge that will cause greater injury in falls and provides a certain contrast from the other surfaces of the stair for easier visual location of the start of the tread surface. The 15 mm limit of beveling and maximum radius of curvature at the leading edge of the tread is intended to allow descending foot placement on a surface that does not pitch the foot forward or allow the ball of the foot to slide off the treads and ascending foot placement to slide onto the tread without catching on a square edge. This section also states that risers shall be solid; however, Section 1011.5.5.3 specifically states exceptions to this requirement that are to be applied. The sloping of risers allows the step profile to have a nosing projection without a lip that might cause a foot to catch when dragged up the face of the riser. Such designs are subject to the maximum nosing projection stated in Section 1011.5.5.1 that must be considered when choosing the angle to slope the riser.

**1011.5.5.1 Nosing projection size.** The leading edge (*nosings*) of treads shall project not more than 32 mm beyond the tread below.

❖ A nosing projection allows the descending foot to be placed further forward on the tread and the heel to then clear the nosing of the tread above as it swings down in an arc, landing on a tread that is effectively deeper than if no nosing projection is used. Nosing projections are so common in stair design that they are usually only noticed by users where they are absent since the lack of nose projection can affect one's gait. Treads with vertical risers are allowed with or without a nosing projection. A nosing projection may also be accommodated by slanting the riser under the tread above. The nosing projection is limited to 32 mm maximum. Treads designed with rounding or bevel on the underside would reduce the chance that a user's toe might catch while ascending the stairway (see Commentary Figure 1011.5.5.1).

**1011.5.5.2 Nosing projection uniformity.** *Nosing* projections of the leading edges shall be of uniform size, including the projections of the *nosing's* leading edge of the floor at the top of a *flight*.

❖ See the commentary to Section 1011.5.5.1 .

**1011.5.5.3 Solid risers.** Risers shall be solid.

**Exceptions:**

1. Solid risers are not required for *stairways* that are not required to comply with Section 1009.3 , provided that the opening between treads does not permit the passage of a sphere with a diameter of 100 mm.
2. Solid risers are not required for occupancies in Group I-3 or in Group F, H and S occupancies other than areas accessible to the public. The size of the opening in the riser is not restricted.
3. Solid risers are not required for *spiral stairways* constructed in accordance with Section 1011.10 .

❖ The code does not address where a riser could contain openings and still be considered solid.

However, the intent is so that someone would not catch their toe as they moved up the stairway (see the commentary for nosing projections in Section 1011.5.5). It is not the intent to prohibit risers made of grills or other designs where a toe would not catch. Grill stairways are often used in exterior locations to allow for the passage of rain to decrease the chance of accumulation and possible slips and falls.

Exception 1 allows the use of open risers on all stairways that are not part of an accessible means of egress. Where the riser is allowed to be open, the opening is limited to be consistent with the requirements for guards (see Commentary Figure 1011.5.5.3). While not required, the second option shown in Commentary Figure 1011.5.5.3 would limit the possibility of a toe catch as someone moved up the stairways and would be a safer design. The code does not reference ICC A117.1 for stairways because stairways are not part of an accessible route; however, the code and standard provide opening limitations in tread surfaces. Section 1011.7.1 states that tread surface openings shall be of a size that does not permit the passage of a 12.5 mm-diameter sphere.

Exception 2 recognizes that open risers are commonly used for stairs in occupancies such as detention facilities, storage, industrial and high-hazard areas for practical reasons. In detention facilities, open risers provide a greater degree of security and supervision because people cannot effectively conceal themselves behind the stairs. Factories, high-hazard buildings and storage facilities have areas where workers may need the open risers to decrease the chance of spillage, water on the stairs. See Section 1011.7.1 for permitted openings in the treads.

Exception 3 recognizes open risers as necessary for adequate foot placement in spiral stairways. The 100 mm opening limitations of Exception 1 are not applicable to spiral stairways.

**1011.6 Stairway landings.** There shall be a floor or landing at the top and bottom of each stairway. The width of landings, measured perpendicularly to the direction of travel, shall be not less than the width of stairways served. Every landing shall have a minimum depth, measured parallel to the direction of travel, equal to the width of the stairway or 1200 mm, whichever is less. Doors opening onto a landing shall not reduce the landing to less than one-half the required width. When fully open, the door shall not project more than 175 mm into the required width of a landing. Where *wheelchair spaces* are required on the *stairway* landing in accordance with Section 1009.6.3 , the *wheelchair space* shall not be located in the required width of the landing and doors shall not swing over the *wheelchair spaces*.

**Exception:**

1. Where *stairways* connect stepped *aisles* to cross *aisles* or concourses, *stairway* landings are not required at the transition between *stairways* and stepped *aisles* constructed in accordance with Section 1030.
2. Where curved stairways of constant radius have intermediate landings, the landing depth shall be measured horizontally between the intersection of the walkline of the lower flight at the landing nosing and the intersection of the walkline of the upper flight at the nosing of the lowest tread of the upper flight.
3. Where a landing turns 90 degrees or more, the minimum landing depth in accordance with this section shall not be required where the landing provided is not less than that described by an arc with a radius equal to the width of the flight served.

❖ A level portion of a stairway provides users with a place to rest in their ascent or descent, to enter a stairway and to adjust their gait before continuing. Landings also break up the run of a stairway, especially at a turn, to aid in the arrest of falls that may occur (see Section 1011.8 ).

The minimum size (width and depth) of all landings in a stairway is determined by the actual width of the stairway. If Section 1011.2 requires a stairway to have a width of at least 1100 mm and the stairway is constructed with that minimum width, then all landings serving that stairway must be at least 1100 mm wide and 1100 mm deep [see Commentary Figure 1011.6(1)]. If a stairway is constructed wider than required, landings must increase accordingly so as to not create a bottleneck situation in the egress travel. However, when a stairway is configured so that it has a straight run, the depth of the landing between flights in the direction of egress travel is not required to exceed 1200 mm, even though the actual width of the stair may exceed 1200 mm [see Commentary Figure 1011.6(2)].

The last portion of the requirement limits the extent to which doors that swing onto landings may interfere or encroach upon the required landing space. This limits the arc of the door

swing on a landing, so that the effect on the means of egress is minimized [see Commentary Figure 1011.6(3)]. This is consistent with a door opening into an exit access corridor in Section 1005.7 . For safety reasons and to ensure the means of egress is continually available for everyone, where an area of refuge/wheelchair space must be located on a landing, the wheelchair spaces must not be within the required landing width and the entrance door to the stair enclosure may not swing over the wheelchair spaces [see Commentary Figure 1009.3.3(1)].

Exception 1 provides a practical exception where assembly facilities are designed for viewing. See Section 1030.14 for assembly stepped aisle walking surfaces. This exception is limited to where stairways are a direct continuation of the path of travel from the level cross aisle to the stepped aisles. It is not permitted for other stairways within the assembly space.

Exception 2 provides criteria for a curved stairway landing with a landing separating two flights aligned in the paths of travel on the stairway run [see Commentary Figure 1011.5.4 and Figure 1011.6(4)]. Due to the tapered shape of the landing, similar to the treads of the adjoining flights, it is logical to regulate the depth at the walkline.

Exception 3 clarifies that it this section is not intended to require that a stairway landing be shaped as a square or rectangle. A landing turning the stairway 90 degrees or more with a curved or segmented outside periphery would be permitted, as long as the landing provides an area described by an arc with a radius equal to the actual stairway width [see Commentary Figure 1011.6(5) and Figure 1011.6(6)]. In this configuration, the space necessary for means of egress will be available on the landings. Where a standpipe is required on these stairways, the standpipes must be located outside of the path of travel per Section 1003.3 .

**1011.7 Stairway construction.** Stairways shall be built of materials consistent with the types permitted for the type of construction of the building.

**Exceptions:**

1 Wood handrails shall be permitted in all types of construction.

2 Interior exit stairways in accordance with Section 510.2 .

❖ In keeping with the different levels of fire protection provided by each of the five basic types of construction designated in Chapter 6, the materials used for stairway construction must meet the appropriate combustibility/noncombustibility requirements indicated in Section 602 for the particular type of construction of the building in which the stairway is located. This is required regardless of whether the stairway is part of the required means of egress. Any structure supporting the stairway and the stairway enclosure must be fire-resistance rated consistent with the construction type; however, the stairway components inside the enclosure need only comply with the material limits for the type of construction.

Per Exception 1, if desired, wood handrails may be used on the basis that the fuel load contributed by this combustible component of stairway construction is insignificant and will not pose a fire hazard.

Exception 2 is a pointer to the specific allowances in Section 510.2 for stairways in buildings with a horizontal separation so that the types of construction on the upper floors could be different from the lower floors and the same stairway moves through both types of construction..

**1011.7.1 Stairway walking surface.** The walking surface of treads and landings of a *stairway* shall not be sloped steeper than one unit vertical in 48 units horizontal (2-percent slope) in any direction. *Stairway* treads and landings shall have a solid surface. Finish floor surfaces shall be securely attached.

**Exceptions:**

1 Openings in *stair* walking surfaces shall be a size that does not permit the passage of 12.5 mm sphere. Elongated openings shall be placed so that the long dimension is perpendicular to the direction of travel.

2 In Group F, H and S occupancies, other than areas of parking structures accessible to the public, openings in treads and landings shall not be prohibited provided a sphere with a diameter of 30 mm cannot pass through the opening.

❖ It is the intent of this section that both landing and stair treads be solid and level with firmly attached surface materials; however, the 1:48 slope should be adequate to allow for drainage to limit the chance for an accumulation of water where someone might slip.

The exceptions permit the use of open grate-type material or slotted grill for stairway treads and landings in two different situations.

Exception 1 allows for a maximum 12.5 mm opening on stairway treads in public areas and serving any use (see Commentary Figure 1011.7.1). This is very beneficial on exterior stairways where water may accumulate. The 12.5 mm limitation is based on the size of a crutch or cane tip and is consistent with ICC A117.1. The opening limitation is also small enough that most shoe heels will not get stuck. If a slotted grill pattern is used, the slots must run side to side on the stairway tread, not nosing to back.

Exception 2 is applicable in factory, industrial, storage and high-hazard occupancies. This provision is intended to apply primarily to stairs that provide access to areas not required to be accessible, such as pits, catwalks, tanks, equipment platforms, roofs or mezzanines. Walking surfaces with limited-size openings are typically used because open grate-type material is less susceptible to accumulation of dirt, debris or moisture, as well as being more resistant to corrosion. Most commercially available grate material is manufactured with a maximum nominal 25 mm opening; therefore, the limitation that the openings not allow the passage of a sphere of 30 mm diameter allows the use of most material as well as accounting for manufacturing tolerances.

The allowances for openings in risers is addressed in Section 1011.5.5.3 .

**1011.7.2 Outdoor conditions.** Outdoor *stairways* and outdoor approaches to *stairways* shall be designed so that water will not accumulate on walking surfaces.

❖ Outdoor stairways and approaches to stairways are to be constructed with a slope that complies with Section 1011.7.1 or are required to be protected such that walking surfaces do not accumulate water. While not specifically stated, any interior locations, such as near a pool, should also have the stair designed to limit the accumulation of water in order to maintain slip resistance (see Section 1003.4 ).

**1011.7.3 Enclosures under interior stairways.** The walls and soffits within enclosed usable spaces under enclosed and unenclosed stairways shall be protected by 1-hour fire-resistance- rated construction or the fire-resistance rating of the stairway enclosure, whichever is greater. Access to the enclosed space shall not be directly from within the stairway enclosure.

**Exception:** Spaces under *stairways* serving and contained within a single residential dwelling unit in Group R-2 or R-3 shall be permitted to be protected on the enclosed side with 12.5 mm gypsum board.

❖ This section addresses the fire hazard of storage under an interior stairway, whether it is an exit access stairway or exit stairway. The stairway must be protected from a storage area under it, even if the stair is not required to be enclosed. The section also requires that the storage area not open into a stairway enclosure. This limits the potential of a fire that starts in the storage area from affecting the means of egress. The exception provides specific criteria for separation for storage areas under an interior stairway for the indicated residential occupancies.

**1011.7.4 Enclosures under exterior stairways.** There shall not be enclosed usable space under *exterior exit stairways* unless the space is completely enclosed in 1-hour fire-resistance- rated construction. The open space under *exterior stairways* shall not be used for any purpose.

❖ If the space under an exterior stairway is to be used, such as for storage, the area below the stairway must be separated from the stairway with walls and a ceiling with a fire-resistance rating of at least 1 hour. If the space under the exterior stairway is open, it must remain free and clear. A fire occurring in this space would jeopardize the use of the stairway for exiting during an emergency.

**1011.8 Vertical rise.** A flight of stairs shall not have a vertical rise greater than 3600 mm between floor levels or landings.

**Exception:** Spiral stairways used as a means of egress from technical production areas.

❖ Between landings and platforms, the vertical rise is to be measured from one landing walking surface to another (see Commentary Figure 1011.8). The limited height provides a reasonable interval for users with physical limitations to rest on a level surface and also serves to alleviate potential negative psychological effects of long and uninterrupted stairway flights.

The exception allows for spiral stairways that only serve technical production areas to eliminate intermediate landings regardless of height. These stairways typically have limited use and serve areas such as catwalks or lighting booths in stadiums and theaters. In addition, there is the technical difficulty of

maintaining proper headroom in a spiral stairway with an intermediate landing.

**1011.9 Curved stairways.** Curved stairways with winder treads shall have treads and risers in accordance with Section 1011.5 and the smallest radius shall be not less than twice the minimum width or required capacity of the stairway.

**Exception:** The radius restriction shall not apply to curved stairways in Group R-3 and within individual dwelling units in Group R-2.

❖ Curved stairway construction consists of a series of winder treads that form a stairway configuration. Options are many, including circular, S-shaped, oval, elliptical, hourglass, and so on. The commentary to Section 1011.5.3 regarding the possible event of nonconcentric movement on stairways with winders also applies to curved stairways. This type of stairway is allowed to be used as a component of a means of egress where tread and riser dimensions meet the requirements of Section 1011.5. This section also requires that the smallest radius must be equal to or greater than twice the required width (see Section 1011.2) of the stairway to limit the degree of turning, thereby expediting egress from higher occupancies (see Commentary Figure 1011.9).

The exception for residential units eliminates the minimum radius requirement where the occupants are familiar with the extent of the turning of the stair through the curve.

**1011.10 Spiral stairways.** *Spiral stairways* are permitted to be used as a component in the *means of egress* only within *dwelling units* or from a space not more than 23 $m^2$ in area and serving not more than five occupants, or from *technical production areas* in accordance with Section 410.5 . A *spiral stairway* shall have a 170 mm minimum clear tread depth at a point 300 mm from the narrow edge. The risers shall be sufficient to provide a headroom of 2000 mm minimum, but riser height shall not be more than 240 mm. The minimum *stairway* clear width at and below the *handrail* shall be 650 mm.

❖ A spiral stairway shall have a 170 mm minimum clear tread depth at a point 300 mm from the narrow edge. The risers shall be sufficient to provide a headroom of 1975 mm minimum, but riser height shall not be more than 240 mm. The minimum stairway clear width at and below the handrail shall be 650 mm.

Spiral stairways can be used within individual dwelling units, from small spaces in other occupancies and from technical production areas in spaces such as theaters. Spiral stairways are also permitted: where specifically indicated, such as to provide access between the levels within a live/work unit (see Section 508.5.5); within staff locations, such as serving guard towers in jails (Section 408.3.4); and technical production areas (Section 410.5.3.4).

Spiral stairways are generally constructed with a fixed center pole that serves as either the primary or the only means of support from which pie-shaped treads radiate to form a winding stairway. The term "spiral" in the geometrical sense describes a curve that diminishes in radius and relates to the form of the stair as viewed in perspective from above or below; however, "spiral" does not describe the actual geometry of the stairway. The unique turning of spiral stairs allows the center pole to act as the guard at the inside of the stair, and the typically narrow width requires users to choose a walkline along the outer perimeter near the only required handrail. On spiral stairways of larger widths where two users can pass and restrict access to the single handrail, the provision of a handrail at both sides should be considered.

The commentary to Section 1011.5.3 regarding the possible event of nonconcentric movement on stairways with winders also applies to spiral stairways. The nature of stairway construction is such that it does not serve well when used in emergencies that require immediate evacuation, nor does a spiral stairway configuration permit the handling of a large occupant load in an efficient and safe manner. Further-more, it is impossible for fire service personnel to use a spiral stairway at the same time and in a direction opposite that being used by occupants to exit the premises, possibly causing a serious delay in fire-fighting operations. Therefore, this section allows only very limited use of spiral stairways where used as part of a required means of egress. Spiral stairways may be used in any occupancy as long as such stairways are not a component of a required means of egress.

Spiral stairways are required to have dimensional uniformity. The stairway must have a clear width of at least 650 mm at and below the handrail. The depth of the treads must not be less than 170 mm measured at a point that is 300 mm out from the narrow edge (see Commentary Figure 1011.10). Riser heights are required to be the same throughout the stairway but are not to exceed 240 mm. Minimum headroom of 1950 mm is required.

**1011.11 Handrails.** Flights of stairways shall have handrails on each side and shall comply

with Section 1014. Where glass is used to provide the handrail, the handrail shall comply with Section 2407.

**Exceptions**:

1 Flights of stairways within dwelling units and flights of spiral stairways are permitted to have a handrail on one side only.

2 Decks, patios and walkways that have a single change in elevation where the landing depth on each side of the change of elevation is greater than what is required for a landing do not require handrails.

3 In Group R-3 occupancies, a change in elevation consisting of a single riser at an entrance or egress door does not require handrails.

4 Changes in room elevations of three or fewer risers within dwelling units and sleeping units in Group R- 2 and R-3 do not require handrails.

5 Where a platform lift is in a stationary position and the floor of the platform lift serves as the upper landing of a stairway, handrails shall not be required on the stairway, provided that all of the following criteria are met:

6 The stairway contains not more than two risers.

7 A handhold, positioned horizontally or vertically, is located on one side of the stairway adjacent to the top landing.

8 The handhold is located not less than 850 mm and not more than 1050 mm above the bottom landing of the stairway.

9 The handhold gripping surface complies with Section 1014.3 , and is not less than 115 mm in length.

❖ Handrails are required along each side of a flight of stairs; however, handrails are not required along stairway landings. Handrail continuity and extensions that will overlap the landings are addressed in Sections 1014.4 and 1014.6 .

Handrails have four recognized functions in stairway use. First, they serve to guide persons in ascent and descent along the path of egress travel, especially important for those with low vision and in cases of fire where vision might be obscured by smoke. Second, they provide a tool for the user to exert stabilizing forces longitudinally (along the length of the rail), vertically and most importantly, transversely (perpendicular) to the rail as the body transfers weight from side to side with each leg swing of the unique gait used on stairs. Third, they provide for pulling when arms are used to augment legs in ascent of steeper angles or where such climbing strategies result in more efficient use of the strengths of the user. Fourth, they are a tool that can be utilized to help in the arrest of a fall. In these capacities, handrails serve to aid in the use of the stairway and are required on both sides of stairways to allow passing users unencumbered access to a handrail, in compliance with Section 1014. Finally, where glass is the material used to provide the handrail, it must comply with Section 2407.

Note that if the handrail extension is at a location that could be considered a protruding object, the handrail must return to the post at a height of less than 700 mm above the floor. Handrails along the stair flights are not considered protruding objects.

The exceptions state conditions where handrails are required on only one side or are not needed at all. By the nature of their construction, spiral stairways can have only a single handrail (see Exception 1). In accordance with Exceptions 1 and 4, all stairways within dwelling units can have a handrail on one side only, and stairs with three or fewer risers are not required to have any handrails. Since "Stair" is defined as one or more risers, Exceptions 2 and 3 are necessary. Exception 3 exempts the single step at the front or back door of a Group R-3 dwelling unit (i.e., townhouse). Decks, patios and walkways often move down with the grade. Where there are single steps, either off a patio or deck to grade, or along the surface, a handrail is not required [Exception 2, see Commentary Figure 1011.11(1)]. Many of these exceptions dealing with residential units are consistent with SBC 1101/1102.

Exception 5 is unique because it deals with steps to the floor of a platform (wheelchair) lift where the floor of the platform lift is elevated in its parked position. Because the lift moves down to serve as an accessible route, the standard handrails cannot be attached or meet the handrail extensions on the steps. However, these lifts typically are surrounded by short walls that can serve as handholds for someone to grab to stop a possible fall. This condition exists when a platform lift is installed to provide access to a raised area that is so small that the platform lift is a large portion or all of the floor of the space. An example would be a witness

stand in a courtroom where the floor of the lift is the floor of the witness stand, The witness stand is surrounded by low walls. A similar condition can exist for podiums in meeting rooms. This exception provides and describes handholds for this limited situation that assist the person walking up and down the steps and do not interfere with the platform lift operation [see Commentary Figure 1011.11(2)].

For guard requirements at stairways, see Section 1011.13 .

**1011.12 Stairway to roof.** In buildings four or more stories above *grade plane*, one *stairway* shall extend to the roof surface unless the roof has a slope steeper than four units vertical in 12 units horizontal (33-percent slope).

**Exception:** Other than where required by Section 1011.12.1 , in buildings without an occupied roof access to the roof from the top story shall be permitted to be by an *alternating tread device*, a ship's ladder or a permanent ladder.

❖ Because of safety considerations, roofs used for habitable purposes such as roof gardens, observation decks, sporting facilities (including jogging or walking tracks and tennis courts) or similar uses, must be provided with conventional stairways that will serve as required means of egress. Access by ladders or an alternating tread device for such uses is not permitted.

In buildings four or more stories high, roofs that are not used for habitable purposes must be provided with ready access by conventional stairways or by an alternating tread device (see Section 1011.14 ). If this stair is also to provide access to an elevator penthouse on the roof, see additional requirements in Section 1011.12.1 . Two reasons for this are access for roof or roof-top equipment repair and fire department access during a fire event. Sloping roofs with a rise greater than 100 mm for every 300 mm in horizontal measurement (4:12) are exempt from the requirements of this section because of the steepness of the construction and the inherent dangers to life safety.

While it is not specifically required that roof access be through an exit stairway enclosure, since part of the intent is for fire department access to the roof, it is strongly advised. Section 1023.9 requires signage at the level of exit discharge indicating whether the stairway has roof access.

**1011.12.1 Stairway to elevator equipment.** Roofs and penthouses containing elevator equipment that must be accessed for maintenance are required to be accessed by a stairway.

❖ The requirement for a stair to the roof for maintaining elevator equipment correlates the code with ASME A17.1/CSA B44, Safety Code for Elevators and Escalators. This referenced standard (see Section 3001.3 ) has required stairs and a door to access elevator equipment since 1955. More specifically, Section 2.27.3.2.1 of ASME A17.1/CSA B44 states the following: "a stairway with a swinging door and platform at the top level, conforming to 2.7.3.3 shall be provided from the top floor of the building to the roof level. Hatch covers as a means of access to the roofs shall not be permitted." Alternating tread devices or ladders are not permitted as an alternative to the stairway for access to the elevator penthouse. This provision is more specific; therefore, while not prohibiting using the same stairway for access to the roof and the elevator penthouse (see Sections 1011.12 and 1011.12.2 ), access to that elevator penthouse must be via a stairway with door access, not an alternating tread device and hatch.

**1011.12.2 Roof access.** Where a stairway is provided to a roof, access to the roof shall be provided through a penthouse complying with Section 1511.2 .

**Exception:** In buildings without an occupied roof, access to the roof shall be permitted to be a roof hatch or trap door not less than 1.5 $m^2$ in area and having a minimum dimension of 600 mm.

❖ The purpose of the penthouse or stairway bulkhead requirement in this section is to protect the walking surface of the stairway to the roof. The exception provides for situations when roof access is only needed for service or maintenance purposes, and where the access may be permitted by alternatives such as alternating tread devices, ship's ladders or ladders.

**1011.13 Guards.** Guards shall be provided along stairways and landings where required by Section 1015 and shall be constructed in accordance with Section 1015. Where the roof hatch opening providing the required access is located within 3000 mm of the roof edge, such roof access or roof edge shall be protected by guards installed in accordance with Section 1015.

❖ Section 1015.2 requires guards along stairways and landings where the edges of the walking surfaces are 750 mm or more above the floor below. Except where permitted by Section 1015.3 , the top of the guard cannot

also serve as the handrail. The handrail must be located inside the guard because of the differences in height requirements.

While guards are required at the edge of a normally occupied roof by Section 1015, there is also a safety concern for roof areas that need to be accessed by service personnel, inspectors and emergency responders. This requirement for guards provides a minimum measure of safety when the roof access is close to the roof edge. This is consistent with the requirements at mechanical equipment in Section 1015.6 and 1015.7 .

**1011.14 Alternating tread devices.** *Alternating tread devices* are limited to an element of a *means of egress* in buildings of Groups F, H and S from a mezzanine not more than 23 $m^2$ in area and that serves not more than five occupants: in buildings of Group I-3 from a guard tower, observation station or control room not more than 23 $m^2$ in area and for access to unoccupied roofs. *Alternating tread devices* used as a means of egress shall not have a rise greater than 6000 mm between floor levels or landings.

❖ This type of device is constructed in such a way that each tread alternates with each adjacent tread so that the device consists of a system of right-footed and left-footed treads (see Commentary Figure 1011.14).

The use of center stringer construction, half-treads and an incline that is considerably steeper than allowed for ordinary stairway construction makes the alternating tread device unique. However, because of its structural feature, only single-file use of the device (between handrails) is possible, thus preventing the occupants from passing one another. The pace of occupant travel is set by the slowest user, a condition that could become critical in an emergency situation. Furthermore, it is impossible for fire service personnel to use an alternating tread device at the same time and in a direction opposite that being used by occupants to exit the premises, possibly causing a serious delay in fire-fighting operations. For these reasons, this section greatly restricts the use of alternating tread devices as a means of egress.

Alternating tread devices may be used in any occupancy as long as such stairways are not a component of a required means of egress.

Alternating tread devices are considered a modest improvement to ladder construction and, therefore, can be used as an unoccupied roof access in accordance with the requirements of Section 1011.12 .

Alternating tread devices are permitted 6000 mm between landings given their limited application and low occupant loads. In addition, it is recognized a vertical rise higher than the typical stairway is needed for these steeper devices which are used where space is often too restrictive for a regular stairway.

**1011.14.1 Handrails of alternating tread devices.** Handrails shall be provided on both sides of alternating tread devices and shall comply with Section 1014.

❖ For the safety of occupants, this section references the dimensional requirements for handrail locations to be used in conjunction with the special construction features of alternating tread devices provided in Section 1011.14 . Because of the steepness of these devices, handrails on both sides are essential for safe functional use and additional clearances are required so that hand movement will not be encumbered by obstructions. Section 1015.3 , Exception 5, permits the handrails along alternating tread devices to also serve as guards.

**1011.14.2 Treads of alternating tread devices.** *Alternating tread devices* shall have a minimum tread depth of 125 mm, a minimum projected tread depth of 215 mm, a minimum tread width of 175 mm and a maximum riser height of 240 mm. The tread depth shall be measured horizontally between the vertical planes of the foremost projections of adjacent treads. The riser height shall be measured vertically between the leading edges of adjacent treads. The riser height and tread depth provided shall result in an angle of ascent from the horizontal of between 50 and 70 degrees. The initial tread of the device shall begin at the same elevation as the platform, landing or floor surface.

**Exception:** *Alternating tread devices* used as an element of a *means of egress* in buildings from a mezzanine area not more than 23 $m^2$ in area that serves not more than five occupants shall have a minimum tread depth of 75 mm with a minimum projected tread depth of 265 mm. The rise to the next alternating tread surface shall not exceed 200 mm.

❖ Alternating tread stairways are required to have tread depths of at least 125 mm. Tread projections are not to be less than 90 mm when measured from tread nosing to tread nosing [next adjacent tread to the left or right to provide a minimum projected tread depth of 215 mm] (see Commentary Figure 1011.14).

The risers are to be not more than 240 mm when measured from tread to alternating tread (next adjacent tread to the left or right). The rise between treads on the same side would be 480 mm maximum. Applying the limiting dimensions stated above results in a device with a very steep incline that is common to ladders; however, because the device may be walked facing down in descent, it is considered a type of stairway in the code.

Tread widths are required to be a minimum of 175 mm or more. With a center support, the total width will be more than 380 mm.

Although no maximum width of the tread is stated, the device must be of a width to provide for functional use of both handrails at the same time in ascent and descent. For this same reason, handrail heights for alternating tread devices are modified from those stairways in Section 1014.2.

Just using the dimensions could result in an alternating tread device with an angle greater than 75 degrees. In any case, the overall angle of the device must be between 50 and 70 degrees.

For alternating tread devices used as a means of egress from small-area mezzanines as prescribed in the exception, the treads must project at least 190 mm as compared to the 90 mm stated earlier; treads are to be at least 75 mm in depth [compared to 125 mm] and risers are not to exceed 200 mm in height [compared to 240 mm].

**1011.15 Ship's ladders.** Ship's ladders are permitted to be used in Group I-3 as a component of a *means of egress* to and from control rooms or elevated facility observation stations not more than 23 $m^2$ with not more than three occupants and for access to unoccupied roofs. The minimum clear width at and below the *handrails* shall be 500 mm. Ship's ladders shall be designed for the live loads indicated in Section 1607.17 .

❖ Ship's ladders can be used in correctional facilities for access to small control rooms, observation stations and unoccupied roofs. Where approved by the building official, ship's ladders could be used for access to unoccupied roofs in other occupancies. Ship's ladders are of similar gradient or pitch to alternating tread devices; however, the treads span the full width like that of a ladder rather than being staggered to either side (see Commentary Figure 1011.15).

**1011.15.1 Handrails of ship's ladders.** *Handrails* shall be provided on both sides of ship's ladders.

❖ Handrails are needed on both sides to assist in ascent and descent and in the absence of a maximum width. Section 1015.3, Exception 5, permits the handrails along ship's ladders to also serve as guards.

**1011.15.2 Treads of ship's ladders.** Ship's ladders shall have a minimum tread depth of 125 mm. The tread shall be projected such that the total of the tread depth plus the *nosing* projection is not less than 215 mm. The maximum riser height shall be 240 mm.

❖ See Commentary Figure 1011.15 for an example of this configuration.

**1011.16 Ladders.** Permanent ladders shall not serve as a part of the *means of egress* from occupied spaces within a building. Permanent ladders shall be constructed in accordance with Section 306.5 of SBC 501 and designed for the live loads indicated in Section 1607.16 . Permanent ladders shall be permitted to provide access to the following areas:

1. Spaces frequented only by personnel for maintenance, repair or monitoring of equipment.
2. Non-occupiable spaces accessed only by catwalks, crawl spaces, freight elevators or very narrow passageways.
3. Raised areas used primarily for purposes of security, life safety or fire safety including, but not limited to, observation galleries, prison guard towers, fire towers or lifeguard stands.
4. Elevated levels in Group U not open to the general public.
5. Non-occupied roofs that are not required to have *stairway* access in accordance with Section 1011.12.1 .
6. Where permitted to access equipment and appliances in accordance with Section 306.5 of SBC *501*.

❖ Permanent ladders are permitted as a means of ingress and egress to very limited spaces. Typically, these spaces are not considered occupied and, as such, are not required to have a means of egress. While the term "technical production areas" is not used in this section, Section 410.5.3.4, Item 6, allows ladders to be used to access technical production areas.

Item 6 of this section references SBC 501 for where ladders can be used to access mechanical equipment that is located in an elevated space or in a room. The details and construction requirements for a permanent ladder are also found in that section. This will help ensure that permanent ladders are safe and usable, while providing consistency for both the designer and the building official.

## SECTION 1012 RAMPS

**1012.1 Scope.** The provisions of this section shall apply to ramps used as a component of a *means of egress*.

**Exceptions**:

1 Ramped *aisles* within assembly rooms or spaces shall comply with the provisions in Section 1030.

2 Curb ramps shall comply with ICC A117.1.

3 Vehicle ramps in parking garages for pedestrian *exit access* shall not be required to comply with Sections 1012.3 through 1012.10 where they are not an *accessible route* serving *accessible* parking spaces, other required *accessible* elements or part of an accessible *means of egress*.

❖ Ramps provide an alternative method of vertical means of access to or egress from a building. Ramps are required for access to building areas for persons who are mobility impaired (see Chapter 11) and for small changes in floor elevations that are a safety hazard in themselves (see Section 1003.5 ). All ramps intended for pedestrian usage, whether required or otherwise provided, must comply with the requirements of this section. The code considers any walking surface that has a slope steeper than one unit vertical in 20 units horizontal (5-percent slope) to be a ramp (see the definition for "Ramp" in Chapter 2).

As with stairways, it is important to understand the terminology. Exit ramps are ramps that provide a protected path of travel between the exit access and the exit discharge. Interior exit ramps are required to be enclosed in accordance with Section 1023. Exterior exit ramps are protected by the exterior wall of the building and must comply with Section 1027. Exit access ramps are typically unenclosed interior ramps and comply with Section 1019 where they provide access between stories. Exit access travel distance stops at an exit ramp enclosure but includes any travel down an exit access ramp. Ramps that are outside and provide a route from the level of exit discharge to grade are considered part of the exit discharge. See the commentary in Chapter 2 for the defined term, "Exit discharge, level of."

Exception 1 indicates that ramped aisles are addressed in Section 1030. Having this exception at the beginning of the ramp section negates the need for repeated exceptions throughout the ramp provisions. While ramps and ramped aisles may look similar, configurations and how occupants move on and off those walking surfaces are very different. Occupants leave and join ramped aisles along the entire run, while occupants enter the ramp only at the top and bottom. Ramped aisles may have no or only one handrail in order to allow for access to the seats, while most ramps have handrails on both sides. Ramped aisles can have steeper slopes to allow for seating bowls to address line of sight. Section 1030 should be used for ramped aisles between and immediately adjacent to seating or where the ramps are a direct continuation of the ramped aisles and lead to a level cross aisle or floor. Section 1012 is used for ramps that lead from the balcony, concourse or cross aisle to a floor level above or below the seating areas.

Exception 2 references specific curb cut requirements found in Section 406 of ICC A117.1. It is important to realize there are different provisions for curb ramps and ramps. For example, a curb ramp can have a rise of any height and not require handrails. Ramps require handrails where the rise is more than 150 mm.

Exception 3 addresses parking garages. An accessible route is required to and from any accessible parking space, and all ramp provisions must be followed. However, ramps that provide access to and from nonaccessible spaces in the remainder of the parking garage need only comply with the provisions for slope and guard requirements. This permits nonaccessible portions of garages to be constructed as a continuous slope. Ramps that are strictly for vehicles, such as jump ramps, are not required to meet any of the ramp provisions.

**1012.2 Slope.** Ramps used as part of a means of egress shall have a running slope not steeper than 1 unit vertical in 12 units horizontal (8.3-percent slope). The slope of other pedestrian ramps shall not be steeper than 1 unit vertical in 8 units horizontal (12.5-percent slope).

❖ Maximum slope is limited to facilitate the ease of ascent and to control the descent of persons with or without a mobility impairment. The

maximum slope of a ramp in the direction of travel is limited to 1 unit vertical in 12 units horizontal (1:12) (see Commentary Figure 1012.2). Ramps in existing buildings may be permitted to have a steeper slope at small changes in elevation (see ICC A117.1 Section 405.2). An example of a ramp that is not part of a means of egress, and therefore is allowed to be a maximum slope of 1:8, is a loading dock or delivery ramp where the ramp is not part of any required exit discharge.

**1012.3 Cross slope.** The slope measured perpendicular to the direction of travel of a *ramp* shall not be steeper than one unit vertical in 48 units horizontal (2-percent slope).

❖ The limitation of one unit vertical in 48 units horizontal on the slope across the direction of travel is to prevent a severe cross slope that would pitch a user to one side (see Commentary Figure 1012.2).

**1012.4 Vertical rise.** The rise for any *ramp* run shall be 750 mm maximum.

❖ Because pushing a wheelchair up a ramp requires a great deal of energy, landings must be situated so that a person can rest after each 750 mm elevation change (see Commentary Figure 1012.2).

**1012.5 Minimum dimensions.** The minimum dimensions of *means of egress ramps* shall comply with Sections 1012.5.1 through 1012.5.3 .

❖ These minimum dimension requirements allow the ramp to function as a means of egress and an accessible route.

**1012.5.1 Width and capacity.** The minimum width and required capacity of a *means of egress ramp* shall be not less than that required for *corridors* by Section 1020.3 . The clear width of a *ramp* between *handrails*, if provided, or other permissible projections shall be 900 mm minimum.

❖ The requirements for the width of a means of egress ramp is based on the required minimum width [typically 900 mm] and the capacity based on the occupant load to be served (see Section 1005.3.2). Note that the clear width of 900 mm is required between the handrails and any other obstructions (e.g., handrail supports, curbs) for proper clearance for a person in a wheelchair. This is different from stairways where handrails are permitted to project into the required width. The 900 mm minimum clear width between handrails is consistent with ICC A117.1.

**1012.5.2 Headroom.** The minimum headroom in all parts of the *means of egress ramp* shall be not less than 2000 mm above the finished floor of the ramp run and any intermediate landings. The minimum clearance shall be maintained for the full width of the ramp and landing.

❖ The requirement for headroom on any part of an egress ramp is identical to the requirement of a conventional (non-spiral) stairway (see Section 1011.3 ). General headroom heights along the means of egress are addressed in Section 1003.2 .

**1012.5.3 Restrictions.** *Means of egress ramps* shall not reduce in width in the direction of egress travel. Projections into the required *ramp* and landing width are prohibited. Doors opening onto a landing shall not reduce the clear width to less than 1100 mm.

❖ The purpose of not allowing ramps to reduce in width in the direction of egress travel is to prevent a restriction that would interfere with the flow of occupants out of a facility. This would include ramp landings in accordance with Section 1012.6.2 . Handrails are the only exception in accordance with Sections 1012.5.1 and 1014.8 . Doors that open onto a ramp landing, including those at the top and bottom landings, must not reduce the clear width to less than 1100 mm. This is a more restrictive provision than for corridors that would permit the reduction to one-half the required width (see Section 1005.7 ). Since one of the purposes of a ramp is to accommodate persons with physical disabilities, it must provide the additional clear width for access by those confined to wheelchairs without the interference or potential blockage caused by the swing of a door (see Commentary Figure 1012.2 and Figure 1012.5.3).

**1012.6 Landings.** *Ramps* shall have landings at the bottom and top of each *ramp*, points of turning, entrance, exits and at doors. Landings shall comply with Sections 1012.6.1 through 1012.6.5 .

❖ Landings must be provided to allow users of a ramp to rest on a level floor surface and to adjust to the change in floor surface pitch. Landings are required at the top and bottom of each ramp run (see Commentary Figure 1012.6). In addition, Section 1012.4 requires a landing every 750 mm of vertical rise of the ramp. The requirements for landings allow those occupants of the structure the ability to negotiate all changes in direction and prepare themselves to either ascend or descend the ramp and to rest. If there is a door at the top or the bottom of the ramp, there are additional requirements in Section 1012.5.3 for door

swing over the landing and Section 405 of ICC A117.1 for maneuvering space and turning space at the door.

**1012.6.1 Slope.** Landings shall have a slope not steeper than one unit vertical in 48 units horizontal (2-percent slope) in any direction. Changes in level are not permitted.

❖ Landings must be almost flat. This allows persons confined to a wheelchair to come to a complete stop without having to activate the brake or hold themselves stationary at the landing. The maximum slope or cross slope of the landing in any direction is 1:48 (see Commentary Figure 1012.2). This minimum slope is to allow for drainage to limit the accumulation of water on the landing surface.

**1012.6.2 Width.** The landing width shall be not less than the width of the widest *ramp* run adjoining the landing.

❖ The width of all landings must be consistently as wide as the widths of the ramp runs leading to them. Means of egress ramps cannot be reduced in width in the direction of egress travel. This is also applicable to the landings connecting the ramp runs (see Commentary Figure 1012.6).

**1012.6.3 Length.** The landing length shall be 1500 mm minimum.

**Exceptions:**

1. In Group R-2 and R-3 individual *dwelling* and *sleeping units* that are not required to be *Accessible units*, *Type A units* or *Type B units* in accordance with Section 1108, landings are permitted to be 900 mm minimum.
2. Where the *ramp* is not a part of an *accessible route*, the length of the landing shall not be required to be more than 1200 mm in the direction of travel.

❖ The landings for ramps must be at least 1500 mm long (see Commentary Figure 1012.6). This allows persons confined to wheelchairs a sufficient distance to stop and rest along with any persons who may be assisting them. This requirement is directly applicable to straight-run ramps that may require an intermediate landing at every 750 mm of vertical rise (see Commentary Figure 1012.2). If the landing is also to be used to negotiate a change in the ramp's direction, Section 1012.6.4 is applicable. If a door overlaps the landing, Section 1012.5.3 is applicable.

The exceptions provide for smaller landings in dwelling and sleeping units and other locations where the ramp is not part of an accessible route. Exception 1 is consistent with SBC 1101/1102. Exception 2 would be applicable in areas such as service ramps and ramps serving assembly seating areas that do not contain any wheelchair spaces.

**1012.6.4 Change in direction.** Where changes in direction of travel occur at landings provided between *ramp* runs, the landing shall be 1500 mm by 1500 mm minimum.

**Exception:** In Group R-2 and R-3 individual *dwelling* or *sleeping units* that are not required to be *Accessible units*, *Type A units* or *Type B units* in accordance with Section 1108, landings are permitted to be 900 mm by 900 mm minimum.

❖ Where a change in direction is made in the ramp at a landing, the landing must be a square of at least 1500 mm. This allows the person confined to a wheelchair enough room to move off the sloped surface and then negotiate the turn with minimal effort. The length of the landing may need to exceed 1500 mm to match the widths of the two ramp runs. In any case, the landing would still need to be 1500 mm wide (see Commentary Figures 1012.5.3 and 1012.6). If a door overlaps the landing, Section 1012.5.3 is applicable. It is not the intent of this provision to prohibit curved ramps. As long as the cross slope meets the limitations in Section 1012.3, a curved ramp is permitted.

The exception provides for smaller landings in dwelling and sleeping units where the ramp is not part of an accessible route. This is consistent with requirements in the SBC 1101/1102.

**1012.6.5 Doorways.** Where doorways are located adjacent to a *ramp* landing, maneuvering clearances required by *ICC A117.1* are permitted to overlap the required landing area.

❖ This section specifies that the area required for maneuvering to open the door and the area of the landing are allowed to overlap. It is not necessary to provide the sum of the two area requirements (see Commentary Figure 1012.5.3). The idea is that someone coming up or down the sloped surface would be able to move onto the landing and not be stopped on the sloped surface by someone opening the door. Requirements for maneuvering space and turning space at the top and bottom of ramps are found in Section 405 of ICC A117.1. ICC A117.1 requires a turning space at the top or bottom landing of a ramp where the door may be locked. This allows people to turn around to travel back along the ramp.

**1012.7 Ramp construction.** *Ramps* shall be built of materials consistent with the types permitted for the type of construction of the building, except that wood *handrails* shall be permitted for all types of construction.

❖ Material requirements for the type of construction as required by Section 602 for floors are also the material requirements for ramp construction.

**1012.7.1 Ramp surface.** The surface of *ramps* shall be of slip-resistant materials that are securely attached.

❖ As the pace of exit travel becomes hurried during emergency situations, the probability of slipping on smooth or slick floor surfaces increases. To minimize the hazard, all floor surfaces in the means of egress are required to be slip resistant. The use of hard floor materials with highly polished, glazed, glossy or finely finished surfaces should be avoided. This is consistent with Section 1003.4.

Field testing and uniform enforcement of the concept of slip resistance is not practical. One method used to establish slip resistance is that the static coefficient of friction between leather and the floor surface is greater than 0.5. Laboratory test procedures such as ASTM D2047 can determine the static coefficient of resistance. Bulletin No. 4 entitled "Surfaces" issued by the US Access Board (ATBCB) contains further information regarding slip resistance.

**1012.7.2 Outdoor conditions.** Outdoor *ramps* and outdoor approaches to *ramps* shall be designed so that water will not accumulate on walking surfaces.

❖ Outdoor ramps, landings and the approaches to the ramp must be sloped so that surfaces do not accumulate water so as to provide a safe path of egress travel at all times. While not specifically stated, any interior locations, such as near a pool, should also have the ramps designed to limit the accumulation of water in order to maintain slip resistance (see Sections 1003.4 and 1012.7.1).

**1012.8 Handrails.** *Ramps* with a rise greater than 150 mm shall have *handrails* on both sides. *Handrails* shall comply with Section 1014.

❖ To aid in the use of a ramp, handrails are to be provided. Handrails are intended to provide the user with a graspable surface for guidance and support. All ramps with a vertical rise greater than 150 mm between landings are to be provided with handrails on both sides [see Commentary Figures 1012.8(1) and 1014.2].

General strength requirements for handrails are found in Section 1014 with a reference to Section 1607.8. Note that if the handrail extension is at a location that could be considered a protruding object, the handrail extension must return to the post at a height of less than 700 mm above the floor. Handrails along the ramp runs are not considered protruding objects.

Depending on the configuration of the ramp and the adjacent walking surface, ramps may require a combination of handrails, edge protection and guards. See Commentary Figures 1012.8(1), 1012.8(2), 1012.8(3) and 1012.8(4) for illustrations of some alternatives.

**1012.9 Guards.** *Guards* shall be provided where required by Section 1015 and shall be constructed in accordance with Section 1015.

❖ To protect the user from falls to surfaces below, guards are to be provided where the sides of a ramp or landing are more than 750 mm above the adjacent grade. Guards are to be constructed in accordance with Section 1015, including the minimum height of 1000 mm [see Commentary Figure 1012.8(4)]. Except where permitted by Section 1015.3 , the top of the guard cannot serve as the handrail.

Depending on the configuration of the ramp and the adjacent walking surface, ramps may require a combination of handrails, edge protection and guards. See Commentary Figure 1012.8(1), Figure 1012.8(2), Figure 1012.8(3) and Figure 1012.8(4) for illustrations of some alternatives.

**1012.10 Edge protection.** Edge protection complying with Section 1012.10.1 or 1012.10.2 shall be provided on each side of *ramp* runs and at each side of *ramp* landings.

**Exceptions**:

1. Edge protection is not required on *ramps* that are not required to have *handrails*, provided they have flared sides that comply with the *ICC A117.1* curb ramp provisions.
2. Edge protection is not required on the sides of *ramp* landings serving an adjoining *ramp* run or *stairway*.
3. Edge protection is not required on the sides of *ramp* landings having a vertical drop off of not more than 12.5 mm within 250 mm horizontally of the required landing area.

❖ This section of the code now addresses the comprehensive requirements for edge protection for all ramps. It must be noted that

edge protection is not the same as the requirements for guards. The presence of a guard does not necessarily provide adequate edge protection and the presence of adequate edge protection does not satisfy the requirements for a guard. Edge protection is necessary to prevent the wheels of a wheelchair from leaving the ramp surface or becoming lodged between the edge of the ramp and any adjacent construction. For example, a ramp may be located relatively adjacent to the exterior wall of a building. However, between the ramp edge and the exterior wall, there is a strip of earth for landscape purposes. Without adequate edge protection, persons confined to wheelchairs could possibly have their wheels run off the side of the ramp into the landscape, causing them to tip. These requirements are consistent with Section 405 of ICC A117.1.

Exception 1 allows a ramp to have minimal edge protection as long as its vertical rise is 150 mm or less. The exception is predicated on the ramp not needing any handrails, which is established by the provisions of Section 1012.8. Such a ramp would only need flared sides or returned curbs. Edge protection without handrails or guards could be a tripping hazard for ambulatory persons. For specific details of these types of edge protection, the provisions of Section 406 of ICC A117.1 for curb ramps must be followed.

Exception 2 reiterates that edge protection is not literally required entirely around a ramp landing. Obviously, edge protection is not required along that portion of the landing that directly adjoins a ramp run; it is only required along the edges of the landing with a dropoff (other than steps of ramp runs).

Exception 3 states that edge protection is not required for those sides of a ramp landing directly adjacent to the ground surface that gently slopes away from the edge of the landing. If the grade adjacent to the ramp landing slopes no more than 1/2:10 (which equates to 1:20) away from the landing, additional edge protection is not required. Such a gradual slope would not be detrimental to persons confined to wheelchairs as they negotiate the ramp landing. Note that this exception is limited to landings, not the ramp surface itself. The ramp must meet the edge protection in Section 1012.10.1 or 1012.10.2.

Depending on the configuration of the ramp and the adjacent walking surface, ramps may require a combination of handrails, edge protection and guards. See Commentary Figures 1012.8(1), 1012.8(2), 1012.8(3) and 1012.8(4) for illustrations of some alternatives.

**1012.10.1 Curb, rail, wall or barrier.** A curb, rail, wall or barrier shall be provided to serve as edge protection. A curb shall be not less than 100 mm in height. Barriers shall be constructed so that the barrier prevents the passage of a 100 mm diameter sphere, where any portion of the sphere is within 100 mm of the floor or ground surface.

❖ Edge protection for ramps and ramp landings may be achieved with a built-up curb or other barrier, such as a rail, wall or guard. The barrier must be located near the surface of the ramp and landing such that a 100 mm-diameter sphere cannot pass through any openings. An example of an effective barrier would be the bottom rail of a guard system. If the bottom rail is located less than 100 mm above the ramp and landing surface, edge protection has been provided. If a curb option is used, the curb must be a minimum of 100 mm high. The curb or barrier prevents the wheel of a wheelchair from running off the edge of the surface and provides people with visual disabilities a toe stop at the edge of the walking surface (see Commentary Figure 1012.10.1).

**1012.10.2 Extended floor or ground surface.** The floor or ground surface of the *ramp* run, or landing shall extend 300 mm minimum beyond the inside face of a *handrail* complying with Section 1014.

❖ An alternative to providing some type of barrier at the edge of the ramp (see Section 1012.10.1 ) is to make the ramp surface wider than the handrails provided at either side. The combination of the wider surface and the handrail barrier would assist in preventing a wheelchair or crutch tip from moving very far off the ramp during a temporary slip (see Commentary Figure 1012.10.1).

## SECTION 1013 EXIT SIGNS

**1013.1 Where required.** *Exits* and *exit access* doors shall be marked by an *approved* exit sign readily visible from any direction of egress travel. The path of egress travel to *exits* and within *exits* shall be marked by readily visible exit signs to clearly indicate the direction of egress travel in cases where the *exit* or the path of egress travel is not immediately visible to the occupants. Intervening *means of egress* doors within *exits* shall be marked by exit signs. Exit sign placement shall be such that no point in an *exit access corridor* or *exit passageway* is more than 30 m or the *listed*